

1 Generic transport equation

Some definitions:

$$\tilde{Q} = -G_{\parallel}Ay' + G_{\parallel}By + G_{\parallel}R \quad (1)$$

$$\tilde{Q}_{\rho} := \tilde{Q} - \dot{b}\rho yN \quad (2)$$

$$\kappa = \frac{\rho}{\rho_b} \quad (3)$$

$$\frac{\partial}{\partial t} \frac{y}{\rho} - \rho \dot{b} \frac{\partial y}{\partial \rho} = D(\Psi) + S \quad (4)$$

The ASTRA generic transport equation reads:

$$\begin{aligned} Sy + P &= \frac{1}{G} \frac{\partial}{\partial t} (G\kappa y)_{\rho} + \frac{1}{V} \frac{\partial}{\partial \rho} (VM\tilde{Q}_{\rho}) = \frac{1}{G} \frac{\partial}{\partial t} (G\kappa y)_{\rho} + \frac{1}{V} \frac{\partial}{\partial \rho} (VM\tilde{Q}) - \frac{\dot{b}}{W} \frac{\partial}{\partial \rho} (\rho VMN) \\ &= \frac{1}{G} \frac{\partial}{\partial t} (G\kappa y)_x - x(\dot{r} - \dot{b}) \frac{1}{G} \frac{\partial}{\partial x} (G\kappa y) + \frac{1}{V} \frac{\partial}{\partial \rho} (VM\tilde{Q}) - \frac{\dot{b}}{W} \frac{\partial}{\partial \rho} (\rho VMNy) \end{aligned}$$

Substitung the definitions above, the explicit form reads:

$$\begin{aligned} &\frac{1}{G} \frac{\partial}{\partial t} (G\kappa y)_x + \frac{1}{V} \frac{\partial}{\partial \rho} [VM(-G_{\parallel}Ay' + G_{\parallel}By + G_{\parallel}R)] = \\ &= Sy + P + \frac{\dot{b}}{V} \frac{V}{W} \frac{\partial}{\partial \rho} (VMN\rho y) + (\dot{r} - \dot{b}) \frac{\rho}{V} \frac{V}{G} \frac{\partial}{\partial \rho} (G\kappa y) \end{aligned}$$

In implicit form, the transport equation looks like:

$$\begin{aligned} &\frac{1}{G} \frac{\partial}{\partial t} (G\kappa y)_x + \frac{1}{V} \frac{\partial}{\partial \rho} \left[VM \left(-G_{\parallel}Ay' + G_{\parallel}By + G_{\parallel}R - \dot{b} \frac{V}{W} N\rho y - (\dot{r} - \dot{b}) \rho \frac{\kappa y}{M} \right) \right] = \\ &= Sy + P - (\dot{r} - \dot{b}) \frac{GHy}{V} \frac{\partial}{\partial \rho} R \left(\frac{\rho V}{G} \right) - \dot{b} MN\rho y \frac{\partial}{\partial \rho} \left(\frac{V}{W} \right) \end{aligned}$$

2 Physical meaning of the argument variables

	G	H	M	$G_{\parallel}$	N	V	A	B	S	P	W
Ψ	1	1	1/G	1	$1/\rho$	C_1	G_2	0	0	$j_{bs} + \dots$	$1/\rho$
n	V'	1	$1/V'$	$G_{\parallel}$	V'	V'	Diff	V_{pinch}	SNN	SN	V'
F_j	V'	1	$1/V'$	$G_{\parallel}$	V'	V'	Diff	V_p	SFN	SF	V'
T_e	$V'^{5/3}$	n_e	$625/V'$	$G_{\parallel}/625$	$3/2 \ n_e \ V'^{5/3}$	V'	χ_e	V_{pe}	PET	PE	V'
$U_{\parallel}$	V'	UPS	1	1	1	V'	χ_{ϕ}	V_{pu}	0	Torque	V'